MIL-P-63234(AR)
AMENDMENT 2
8 June 1983
SUPERSEDING
AMENDMENT 1
27 October 1980

MILITARY SPECIFICATION

PROJECTILE, GUIDED, 155MM, XM712
ROLL RATE SENSOR FOR

This amendment forms a part of Military Specification MIL-P-63234(AR), dated 31 August 1979, and is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title: Add "M712" after the designation "XM712". Make this addition wherever "XM712" appears in the entire specification.

PAGE 3

* 3.2.1.4: Delete entire paragraph.

PAGE 4

* 3.2.2.2: Delete entire paragraph and substitute the following:

"3.2.2.2 Vibration.- The RRS shall be capable of withstanding a random vibration environment of 7.6 g rms, $0.04G^2/Hz$ power spectral density, 20 to 2000 Hz. During vibration the rate signal shall not exceed $\pm$ 5VDC and shall not exhibit a signal step change greater than $\pm$ 1 VDC. The RRS shall operate during random vibration of 2.4 g RMS at 20 to 2000 Hz and shall meet the performance during vibration of 3.2.1.3 for required data input."

* 3.2.2.3: Delete entire paragraph and substitute the following:

"3.2.2.3 Relative humidity.- The RRS shall withstand relative humidity exposures of greater than 90 percent at + 68°F to + 149°F."

FSC 1320

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PAGE 6

* 4.4.2.2: Add the following words to end of paragraph:

"Rejected lots shall be dispositioned in accordance with the applicable requirements of MIL-STD-105."

PAGE 7

* 4.5.2.5: Delete entire paragraph.

PAGE 8

* 4.5.3.3: Delete entire paragraph and substitute the following:

"4.5.3.3 Vibration test.- The vibration test shall be performed in two phases, gross operating vibration and performance vibration. Total operating time shall be seven minutes, five minutes for gross operating and two minutes for performance. The input vibration amplitude shall be reduced to zero between the two portions. The RRS shall have all input power removed for a minimum of 10 minutes prior to initiation of the Performance Vibration Test. Acceptance criteria shall be as specified in paragraph 3.2.2.2.

"4.5.3.3.1 Gross operating vibration.- The RRS shall be tested while operating during vibration per MIL-STD-810C, Method 5142, Procedure V, Part 2, Curve AF of Figure 514.2-5 at 7.6 g RMS. Power shall be applied per Figure 1 with the exception of the null store time which shall be 3.0 $\pm$ 2.0 sec instead of 1.6 $\pm$ 0.2 sec.

"4.5.3.3.2 Performance vibration.- The RRS shall be tested while operating with power application per Figure 1 during vibration per MIL-STD-810C, Method 514.2, Procedure V, Part 2, Figure 514.2-5, Curve AF except at 2.4 g RMS."

PAGE 8

* 4.5.3.4: Delete entire paragraph and substitute the following:

"4.5.3.4 Relative humidity test.- The RRS shall be tested in accordance with the humidity environment specified in MIL-STD-810, Method 507.1, Procedure V, one cycle only, and shall be tested at 72 $\pm$ 18°F for zero rate input to the requirement of 3.2.1.3 within 1 hour after humidity exposure."

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PAGE 12

Replace Figure 1 with new Figure 1 attached.

PAGE 13

Replace Figure 2 with new Figure 2 attached.

PAGE 15

Replace Figure 4 with new Figure 4 attached.

PAGE 24

* Table II, Major defect 105: Delete in its entirety.

PAGE 25

* Table III, Para Ref/Insp Method: Delete in its entirety and substitute with the following:

"4.5.3.3, 4.5.2.1 thru 4.5.2.4."

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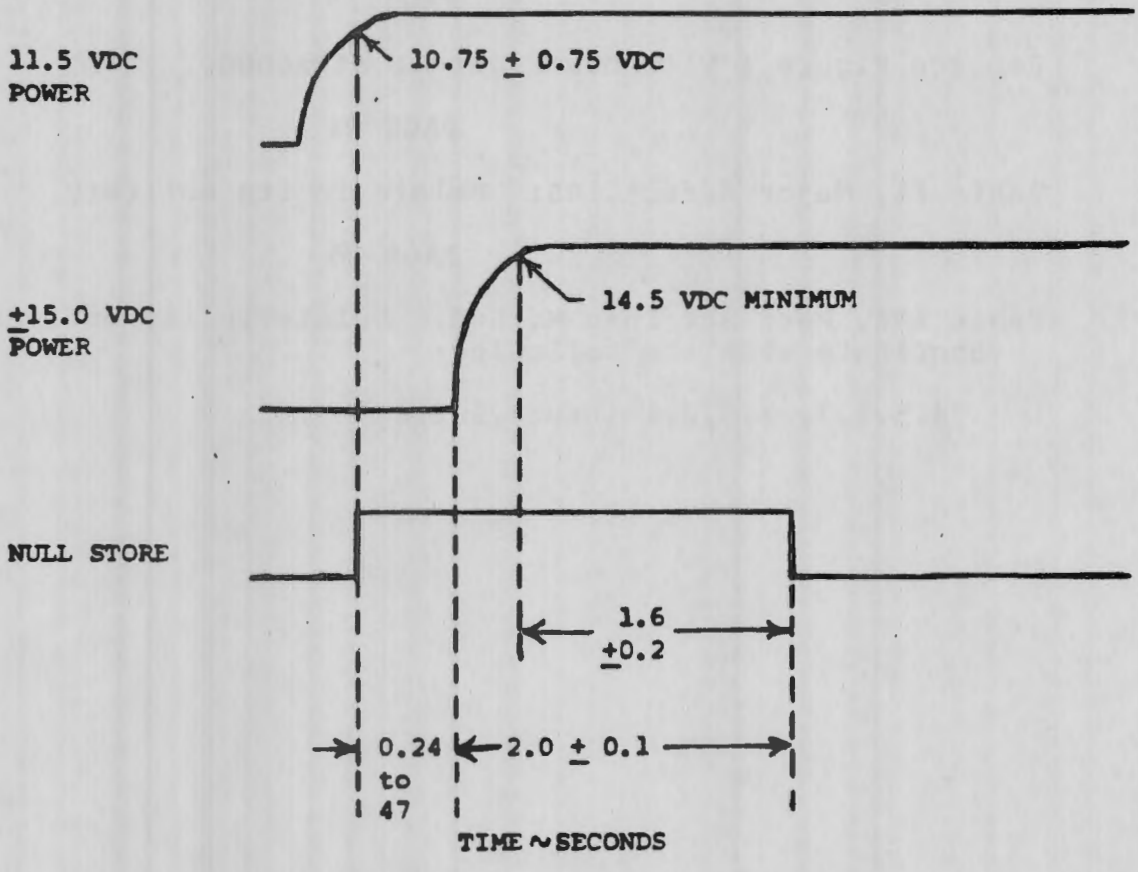


Figure 1. Timing Sequence

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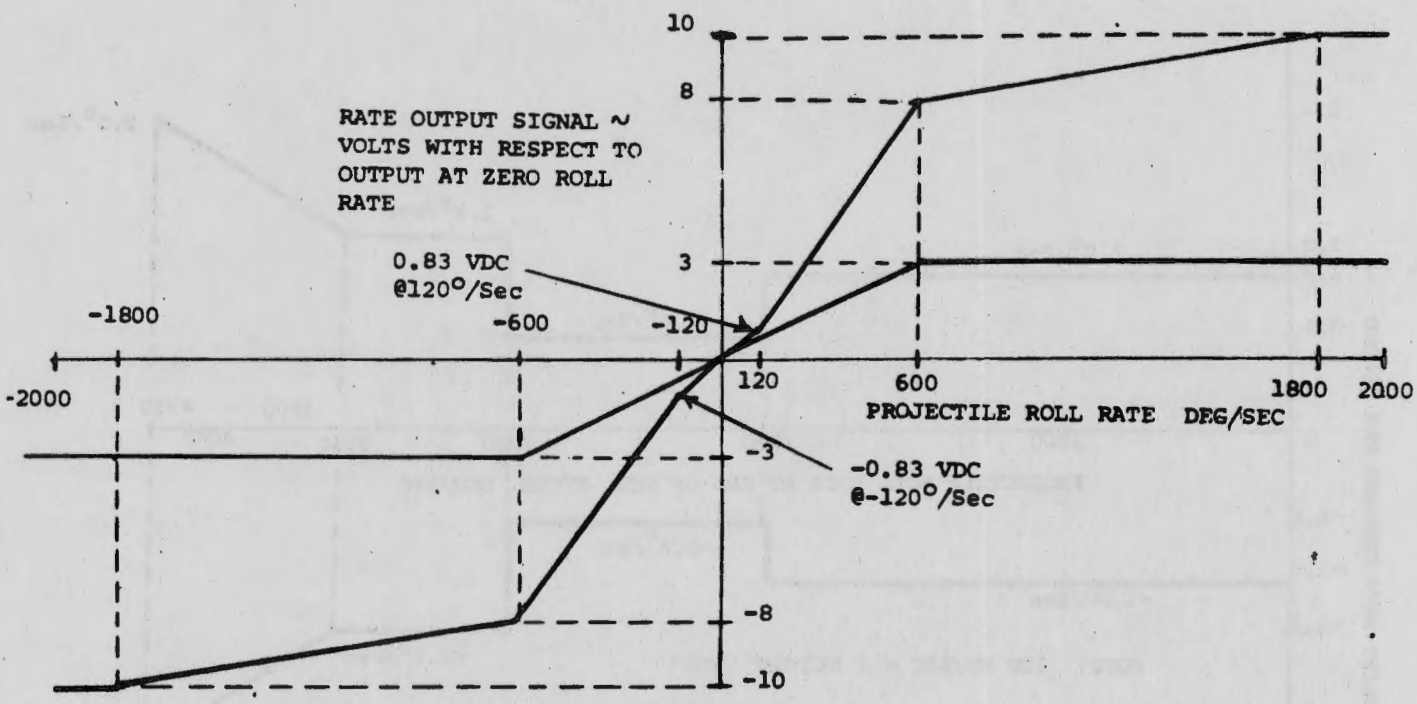


FIGURE 2. Roll Rate Output Boundaries

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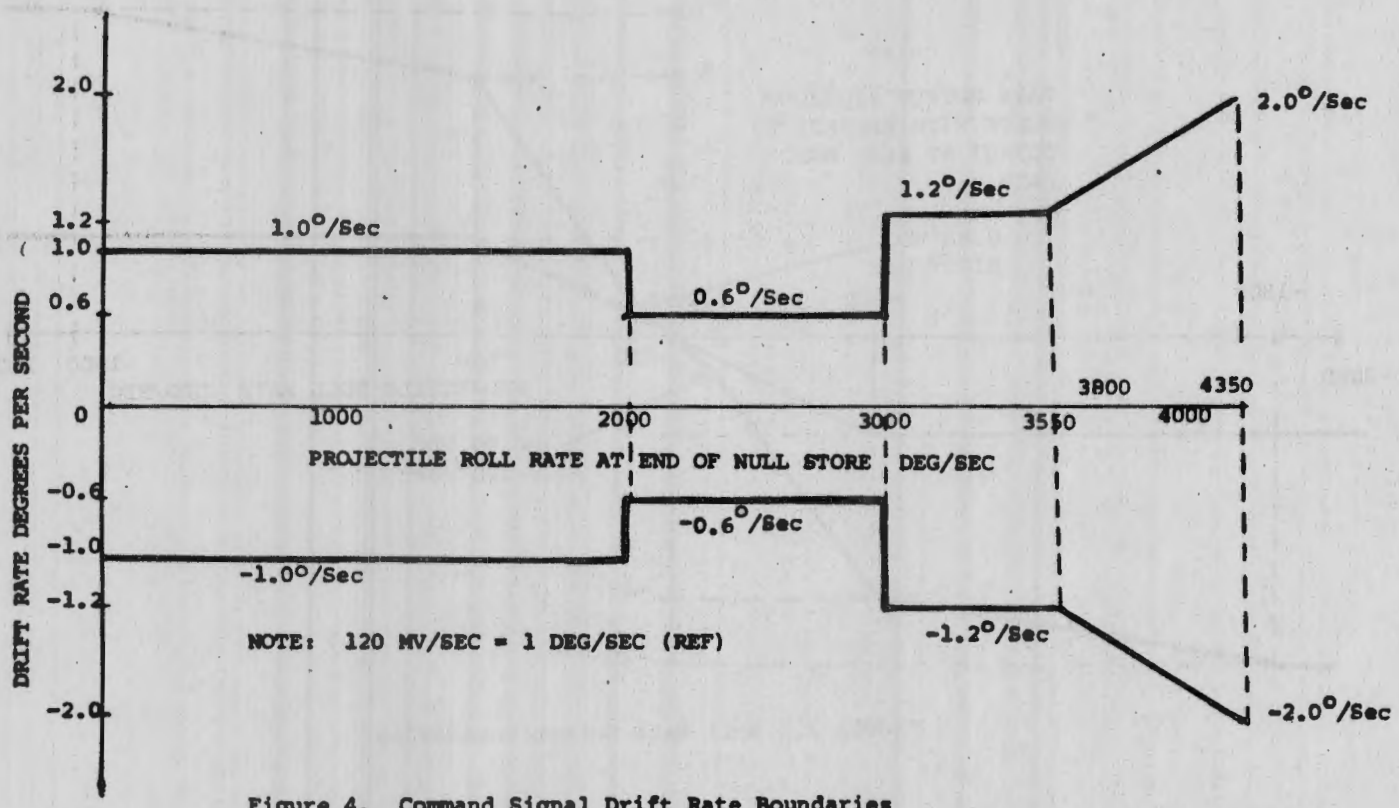


Figure 4. Command Signal Drift Rate Boundaries

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